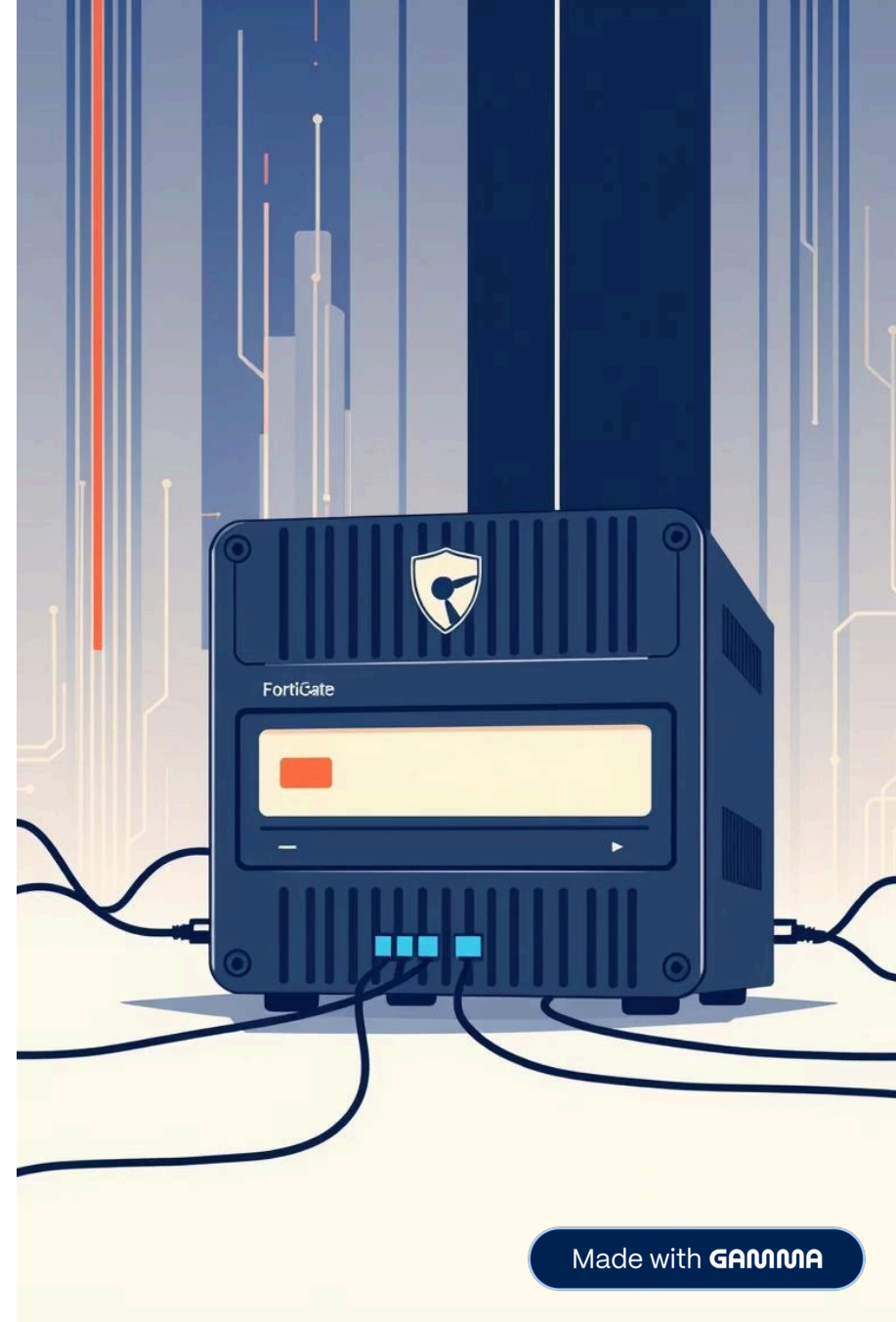


FortiGate VPN Solutions: Project Overview

This presentation outlines a comprehensive 4-week project focused on implementing and optimizing VPN solutions using FortiGate firewalls. Designed for IT network engineers and technical students, this project builds foundational knowledge in VPN concepts and advanced configurations.





Project Goals: Secure Connectivity & Optimization

Master VPN Concepts

Gain a deep understanding of various VPN types and their practical applications in modern networks.

Hands-on Configuration

Develop practical skills in configuring both SSL and IPsec VPNs on FortiGate devices.

Optimize Traffic Flow

Learn to integrate SD-WAN with VPNs for enhanced performance and intelligent traffic routing.

Comprehensive Documentation

Produce detailed guides and reports on configurations, testing, and performance analysis.

Week 1: Understanding & Configuring SSL VPN

1 VPN Concepts & Use Cases

Explore different VPN types, including site-to-site, remote access, and their respective scenarios for secure communication.

2 SSL VPN Deep Dive

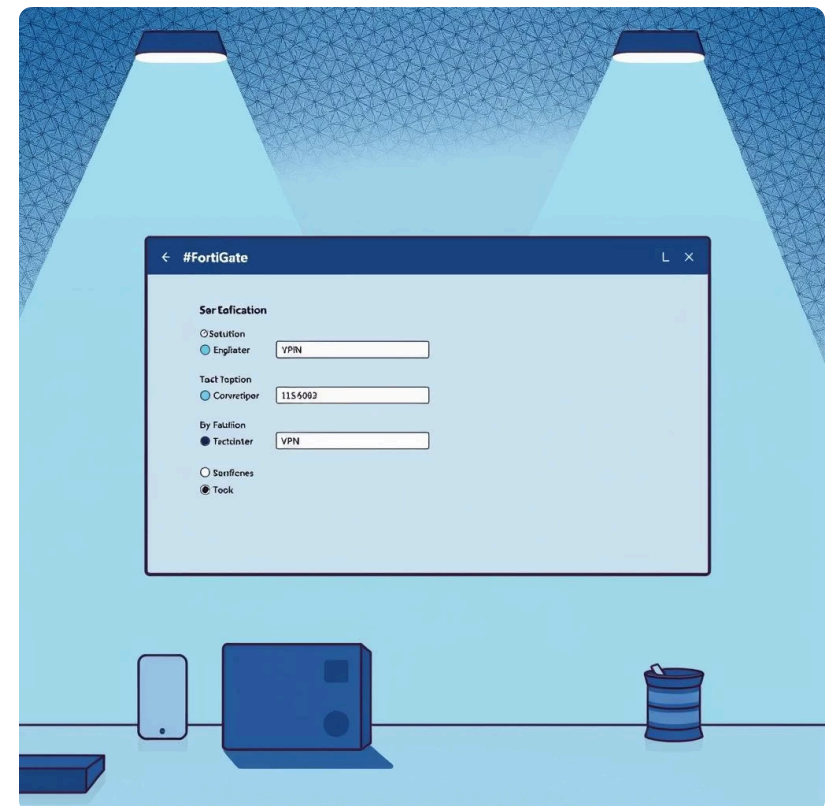
Understand the architecture, authentication methods, and encryption protocols behind SSL VPNs.

3 FortiGate SSL VPN Configuration

Step-by-step guidance on setting up an SSL VPN on a FortiGate device, including user authentication and portal customization.

4 Testing & Troubleshooting

Verify SSL VPN connectivity and learn common troubleshooting techniques for resolving access issues.



Deliverable: Week 1 - SSL VPN Guide



Configuration Guide

A detailed document outlining the configured SSL VPN parameters and settings on the FortiGate.



Test Results

Documentation of successful connection tests from remote clients, including access to internal resources.

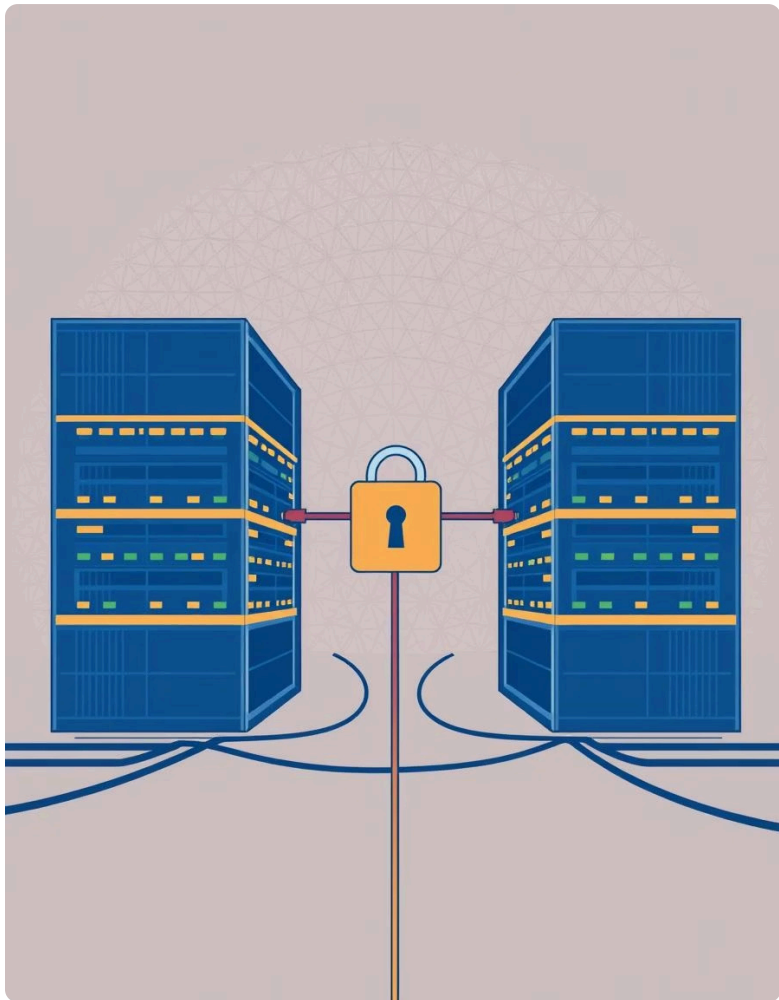


Network Diagram

Visual representation of the SSL VPN topology and connected components.

This deliverable serves as a comprehensive resource for future reference and replication of the SSL VPN setup.

Week 2 & 3: Implementing SDWAN Over IPsec VPN Tunnels



01

SDWAN VPN

- SDWAN Configured by GUI

Network→SDWAN→Create Members

02

Create 2 IPsec Tunnels

- Create Phase 1 and Phase 2 with the remote IPs and configuring Preshared Key

03

Phase 2 Configuration (IPsec)

- Set up the IPsec tunnel to secure data transfer, defining proxies, protocols, encryption algorithms and the 2 networks.

04

Routing & Policies

- Implement static routes and firewall policies to allow traffic to traverse the SDWAN IPsec VPN tunnels securely.

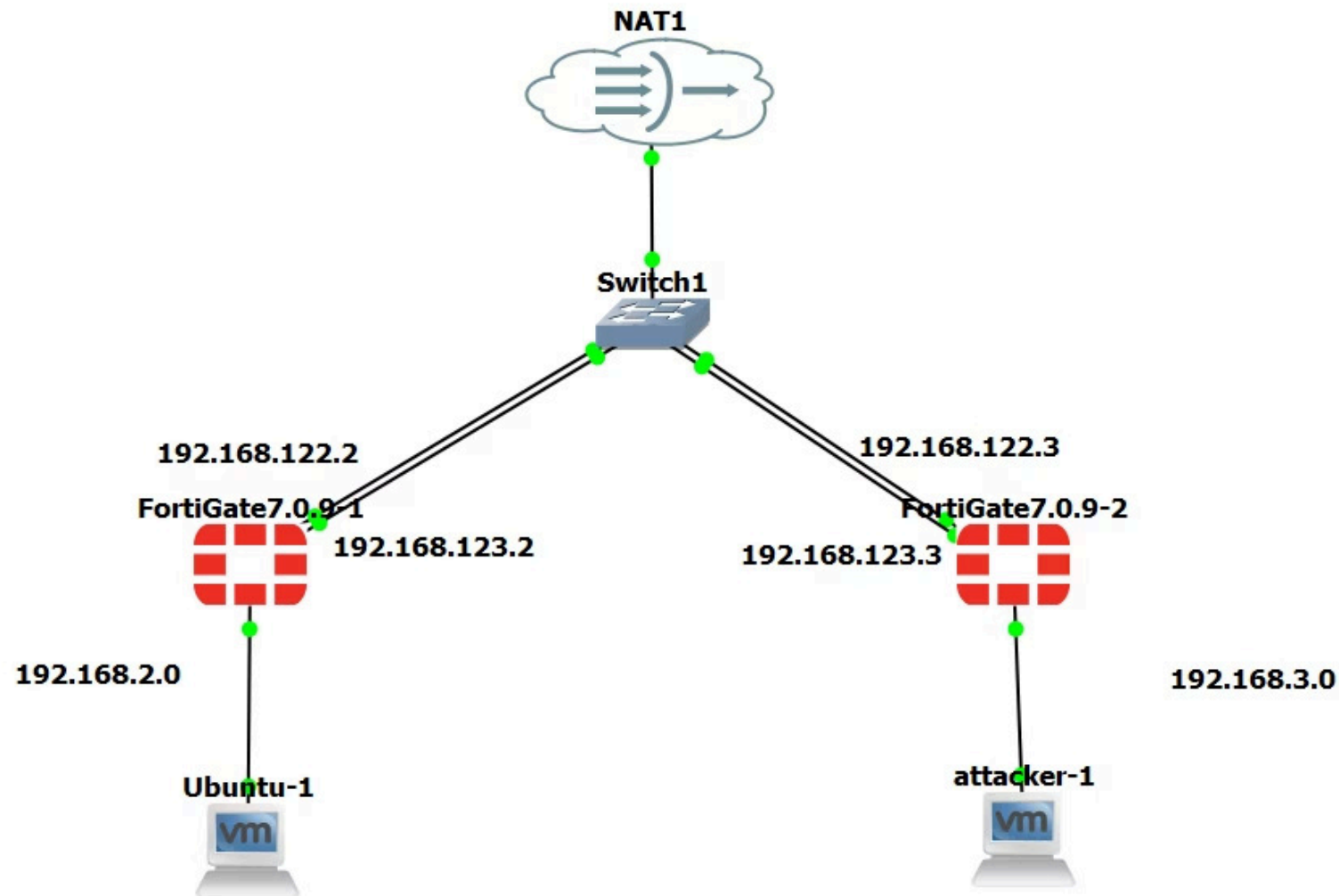
05

Configuring SLA & SDWAN rule

- Implement static routes and firewall policies to allow traffic to traverse the SDWAN IPsec VPN tunnels securely.

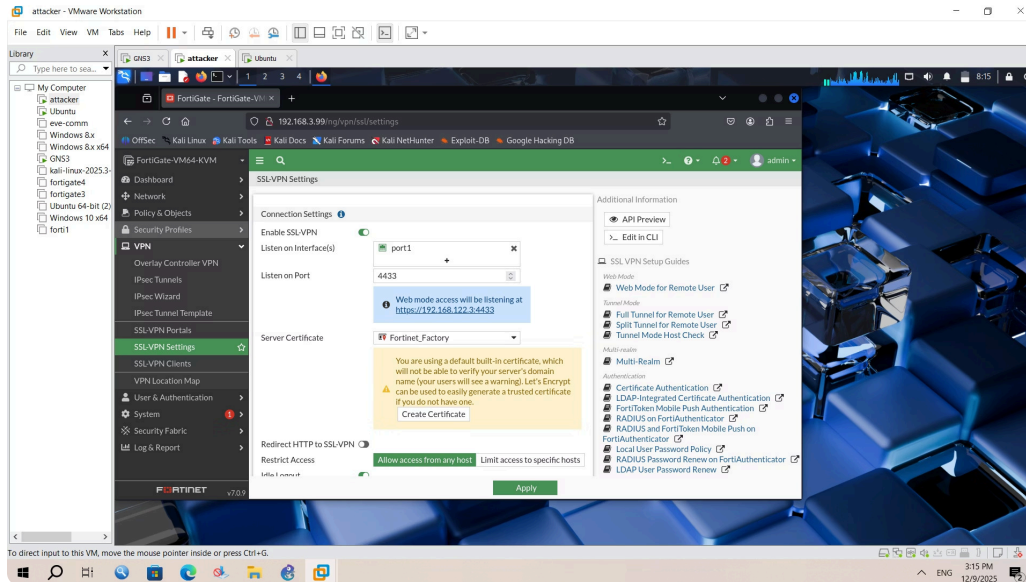
Deliverable: Week 2 & 3 - SDWAN Over IPsec VPN Documentation

→GNS3



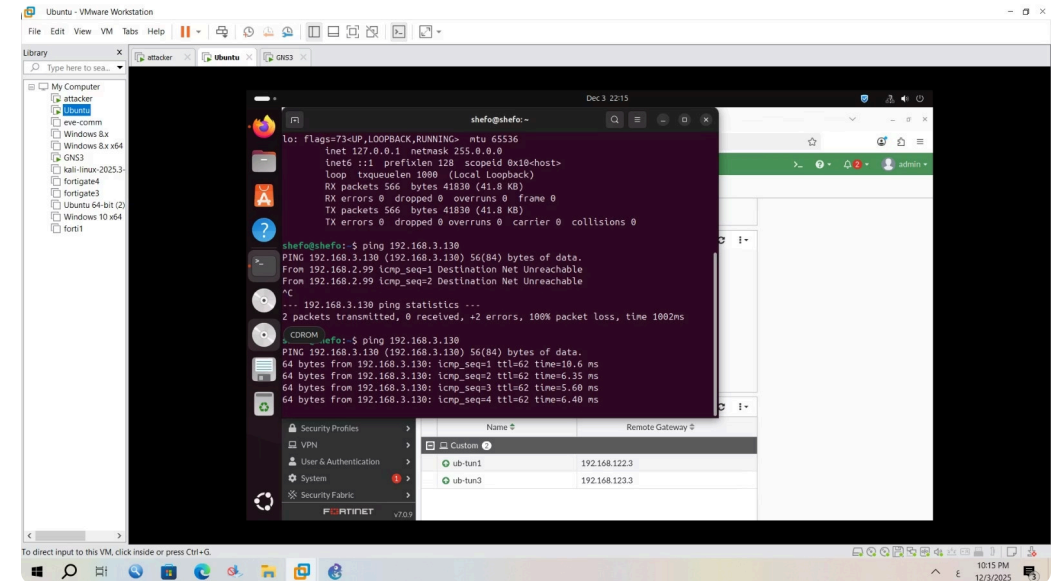
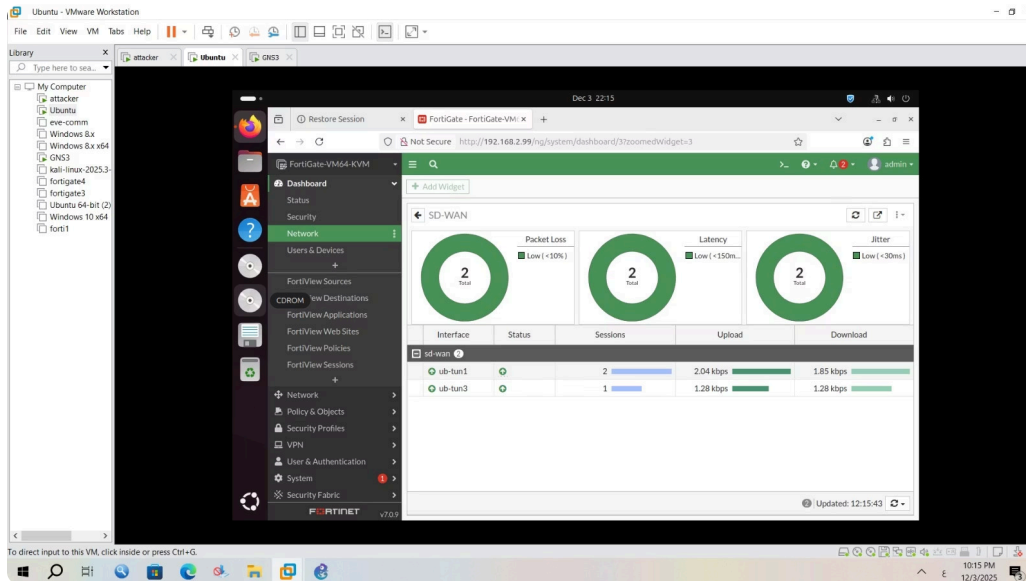
Chapter Testing

Week 1: SSLVPN settings Net LAN 192.168.3.0 (Branch)



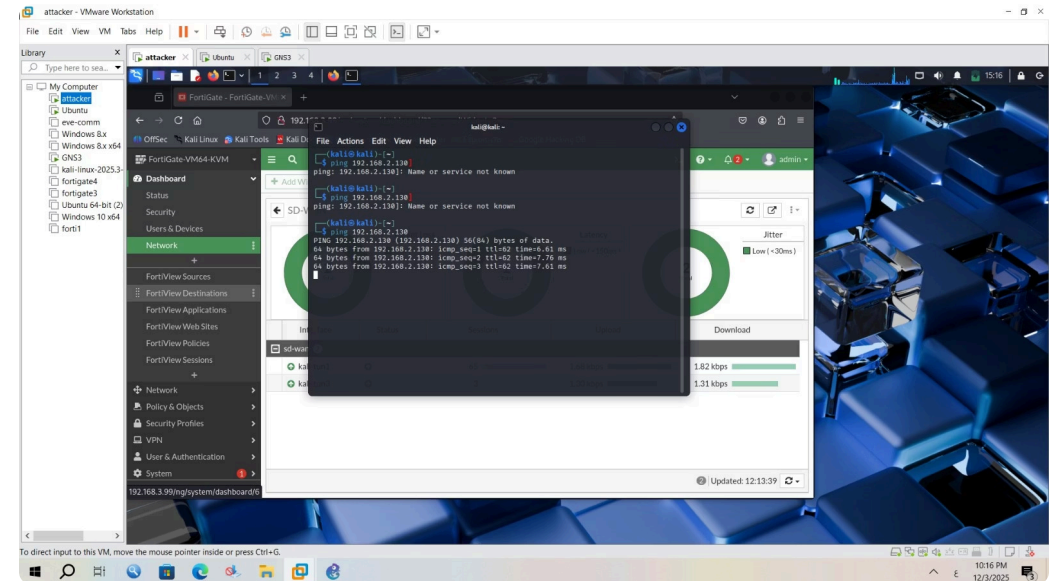
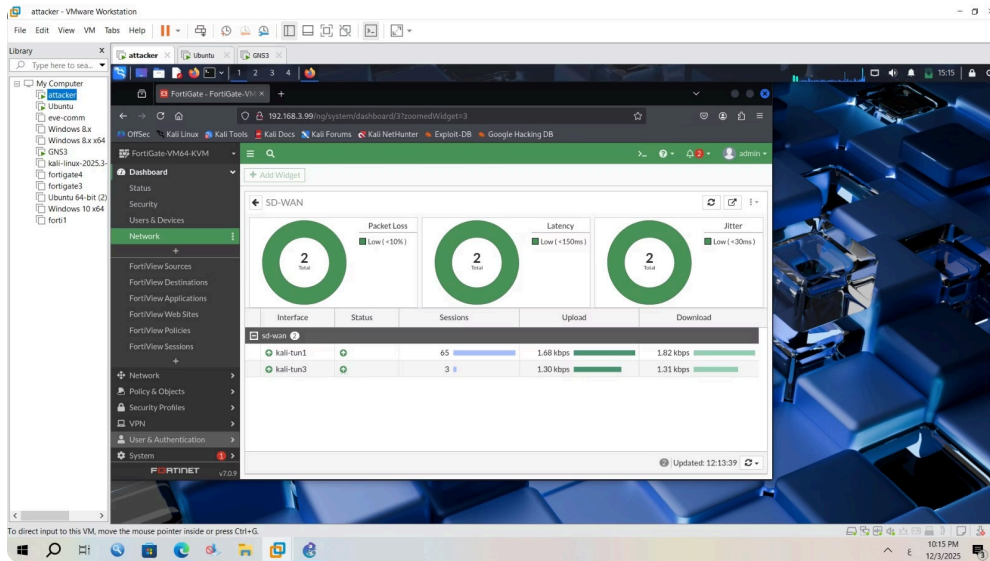
Chapter Testing

Week 2 & 3: Cont Net LAN 192.168.2.0 (HQ)



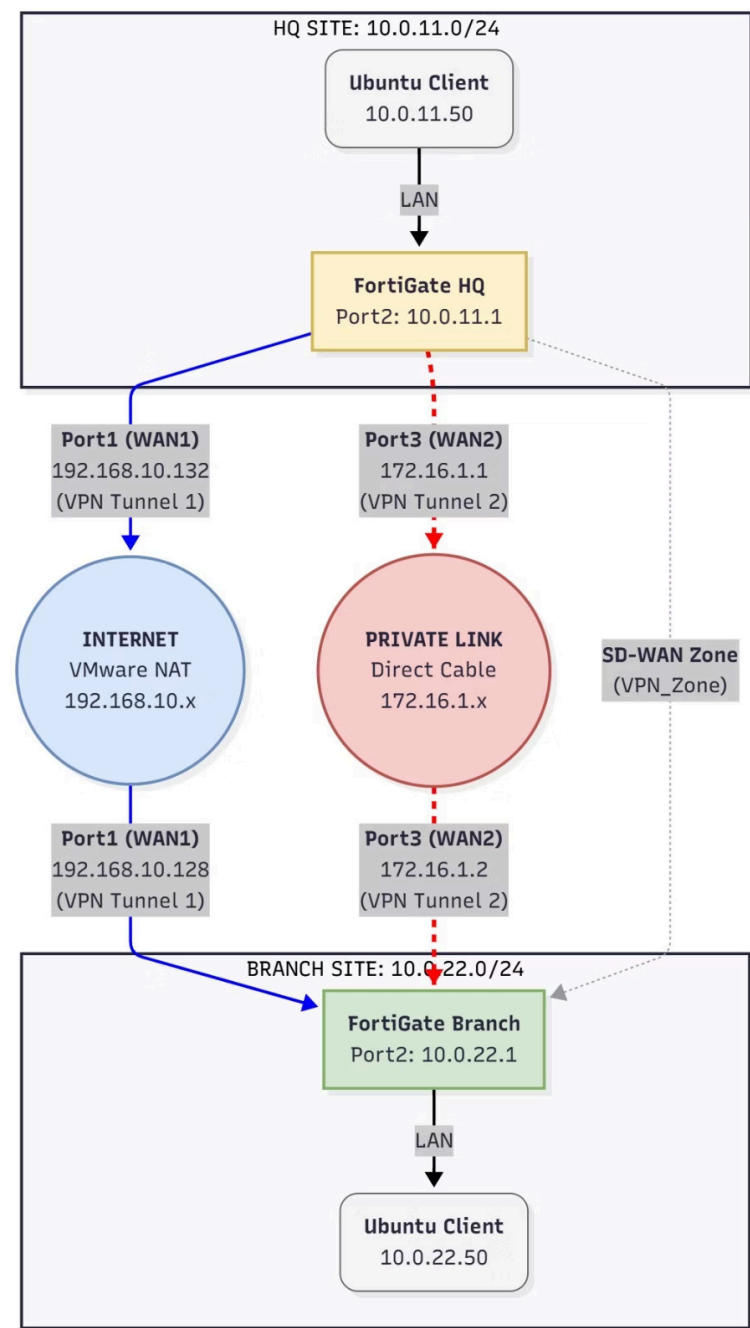
Chapter Testing

Week 2 & 3: Cont Net LAN 192.168.3.0 (Branch)



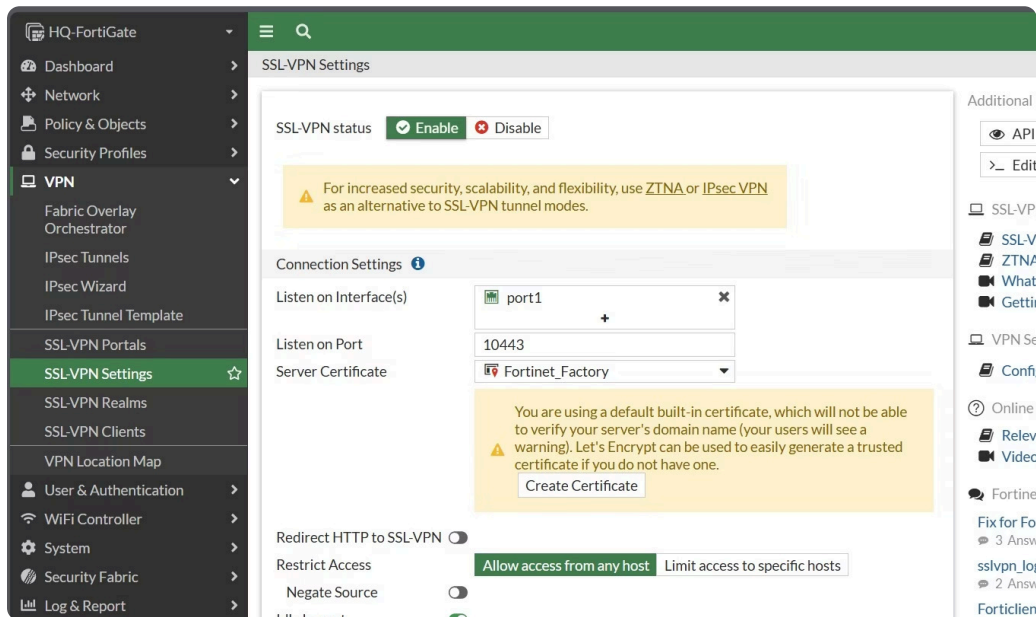
Deliverable: Week 2 & 3 Cont - SDWAN Over IPsec VPN Documentation

→VMWare WorkStation



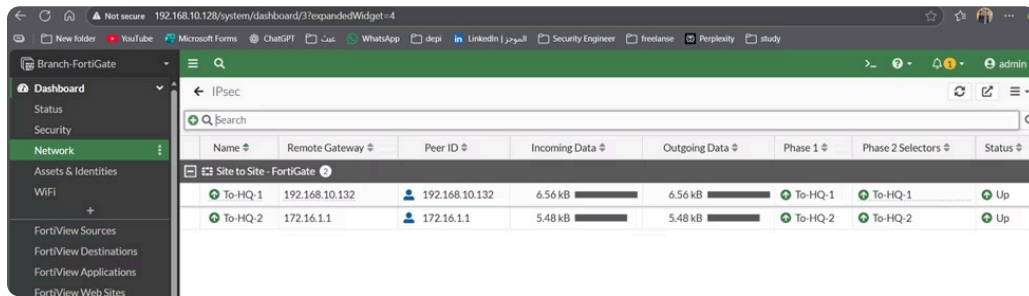
Chapter Testing

Week 1: SSLVPN Net Site HQ Firewall

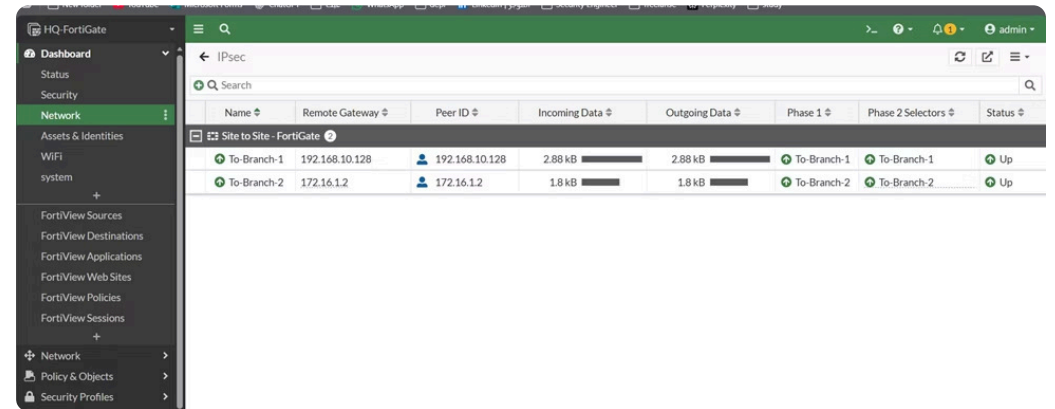


Chapter Testing

Week 2 & 3: IPSec tunnels Net Site HQ & Branch



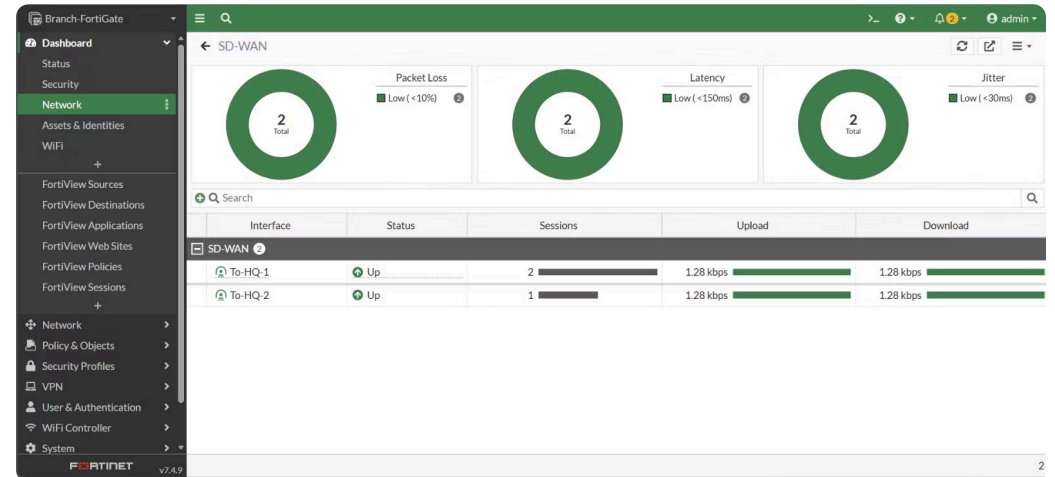
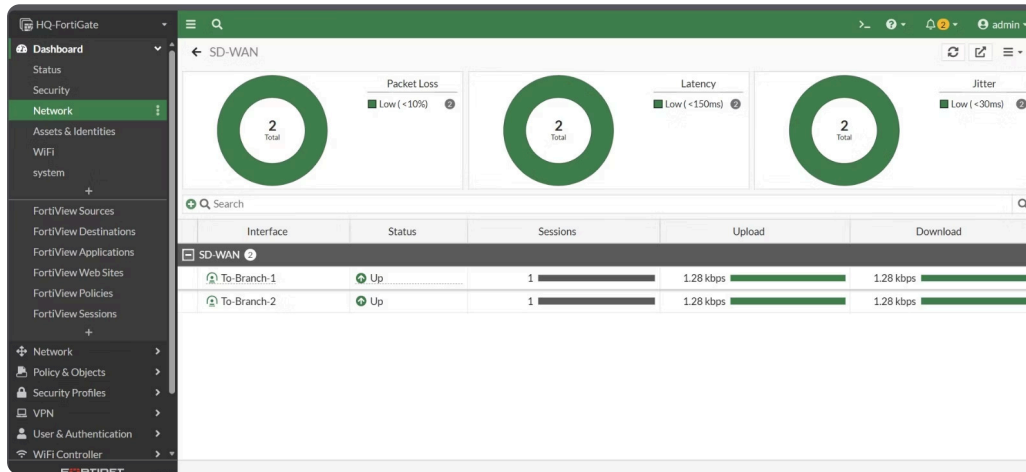
Name	Remote Gateway	Peer ID	Incoming Data	Outgoing Data	Phase 1	Phase 2 Selectors	Status
Site to Site - FortiGate							
To-HQ-1	192.168.10.132	192.168.10.132	6.56 kB	6.56 kB	To-HQ-1	To-HQ-1	Up
To-HQ-2	172.16.1.1	172.16.1.1	5.48 kB	5.48 kB	To-HQ-2	To-HQ-2	Up



Name	Remote Gateway	Peer ID	Incoming Data	Outgoing Data	Phase 1	Phase 2 Selectors	Status
Site to Site - FortiGate							
To-Branch-1	192.168.10.128	192.168.10.128	2.88 kB	2.88 kB	To-Branch-1	To-Branch-1	Up
To-Branch-2	172.16.1.2	172.16.1.2	1.8 kB	1.8 kB	To-Branch-2	To-Branch-2	Up

Chapter Testing

Week 2 & 3: SDWAN Over IPSec Net Site HQ & Branch



Chapter 4: Project Showcase

Week 4: Final Report & Presentation

- 1

Compile All Documentation

Integrate all weekly deliverables into a cohesive final report.
- 2

Executive Summary

Highlight key findings, configurations, and performance improvements.
- 3

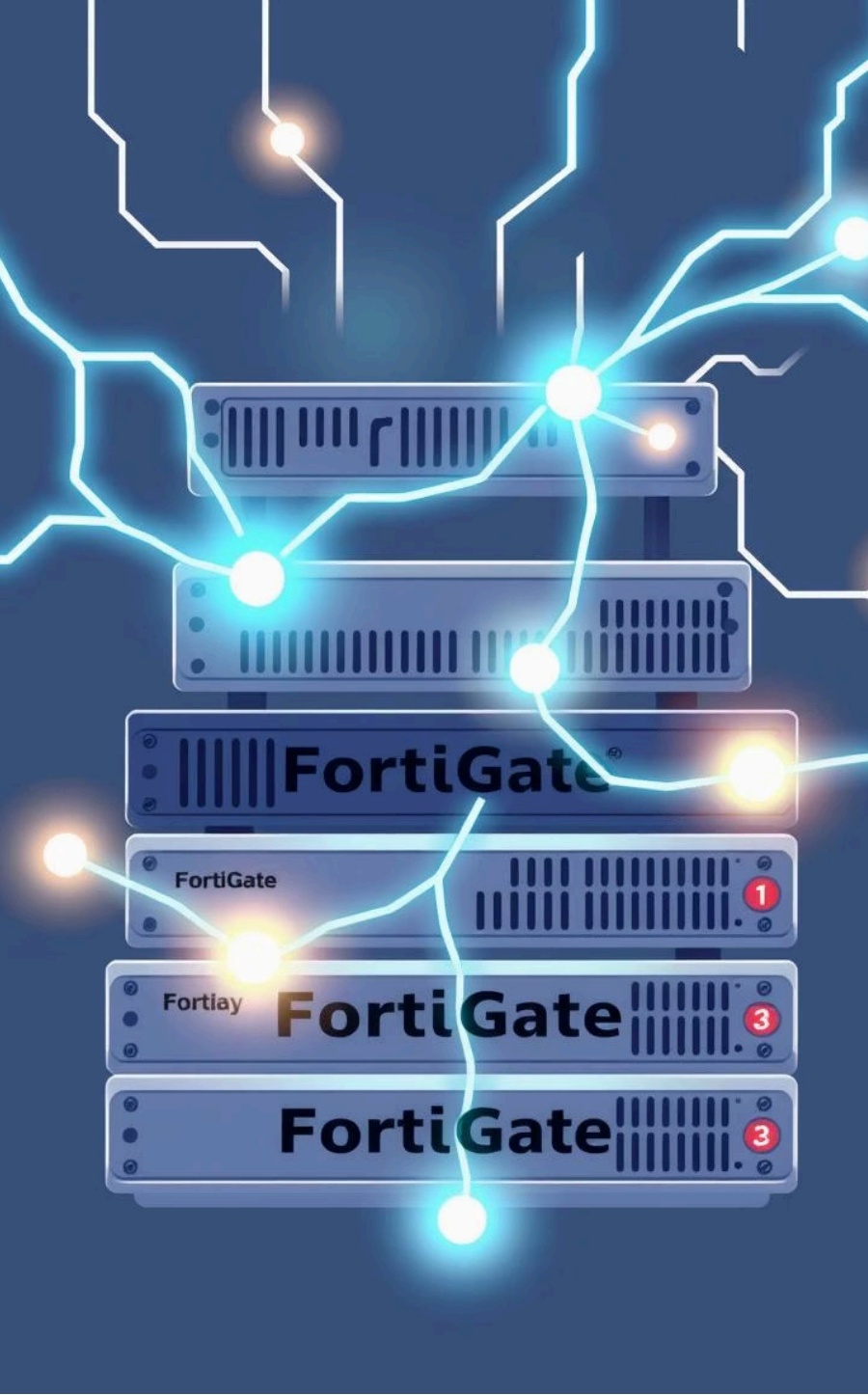
Prepare Presentation

Create a clear and engaging presentation summarizing the project.
- 4

Demonstration

Showcase live VPN and SD-WAN functionality, if applicable.





Key Takeaways & Next Steps



Enhanced Security

FortiGate VPNs provide robust, encrypted communication channels.



Optimized Performance

SD-WAN integration elevates VPN efficiency and user experience.



Continuous Learning

The project offers a strong foundation for advanced network security roles.

This project equips participants with critical skills for deploying and managing secure, high-performance network infrastructures using FortiGate.

Meet Our Team